Capabilities compared with the graph-colouring line of work

	Lee et al. 2026	Jain & Kumar 2026	MKCO (ours)
Commuting circuits	✓	✓	✓
Non-commuting circuits	–	✓	✓
Exact reduction for general circuits	–	–	✓ (Thm 1)
Certified lower bound / optimality proof	partial (Eq. 2)	–	✓ (Thm 7, Prop 9)
Depth budget and soft constraints (crosstalk)	–	–	✓ (Thm 2)
Capacities (ancilla fan-out)	heuristic	–	✓ (capacitated cut)
Quantum-native solver path (Ising)	–	–	✓ (Thm 4, Prop 6)
Memory without an $O(n^2)$ conflict graph	–	–	✓
Independent equivalence audit	–	–	✓
Non-Clifford depth with free Cliffords	partial (commuting)	–	✓ (Prop 10)
Commutation-aware routing	–	–	✓ (Prop 11)
State-aware idle-noise scoring	–	–	✓ (Prop 12)